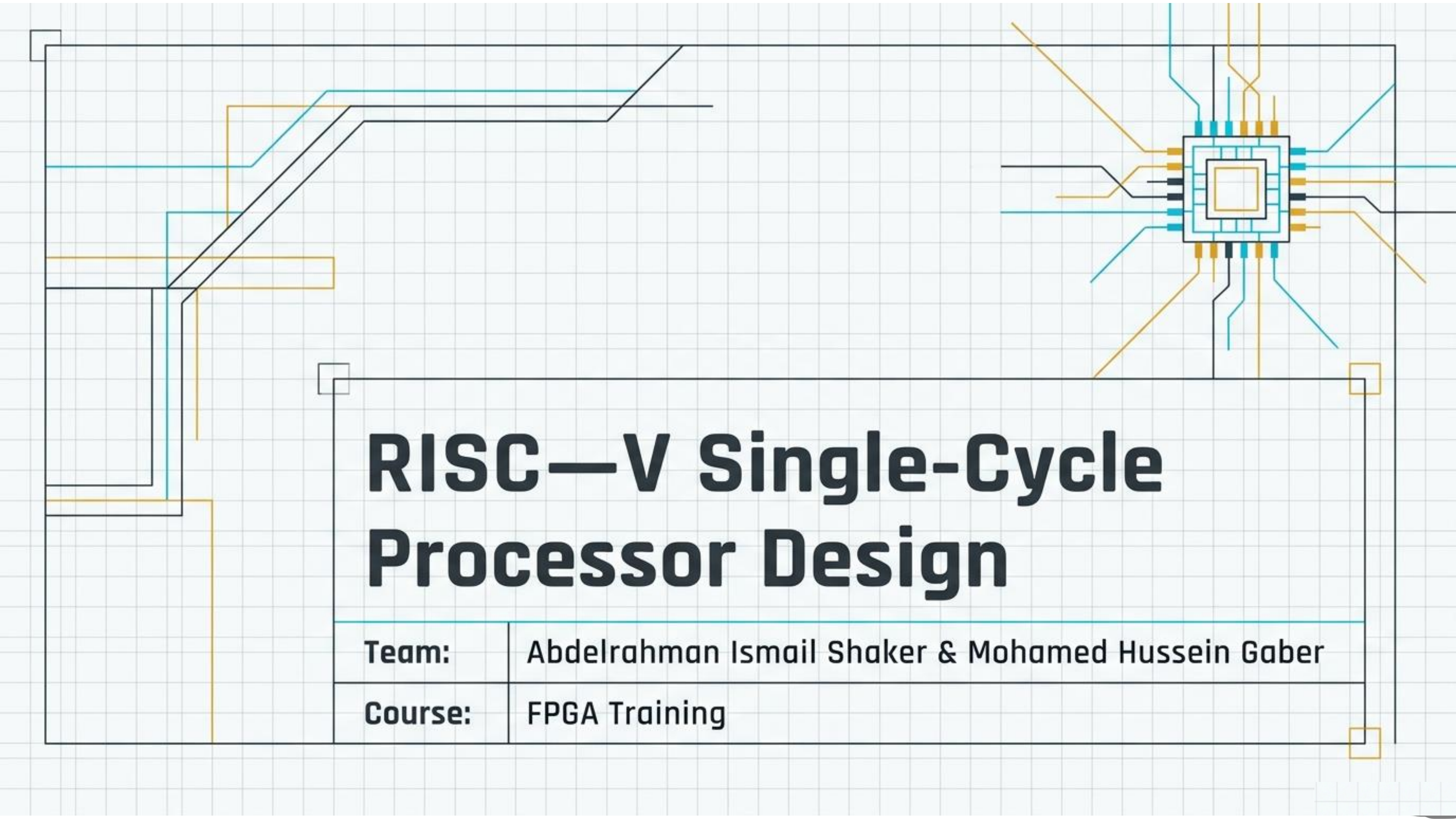
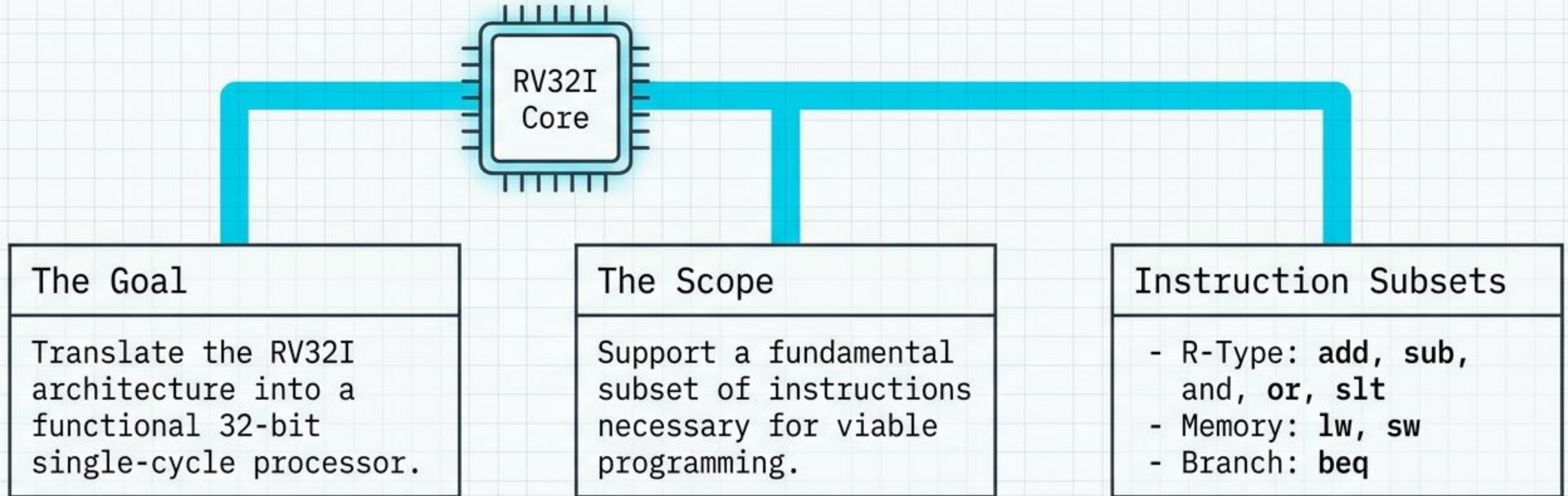




RISC—V Single-Cycle Processor Design

Team:	Abdelrahman Ismail Shaker & Mohamed Hussein Gaber
Course:	FPGA Training

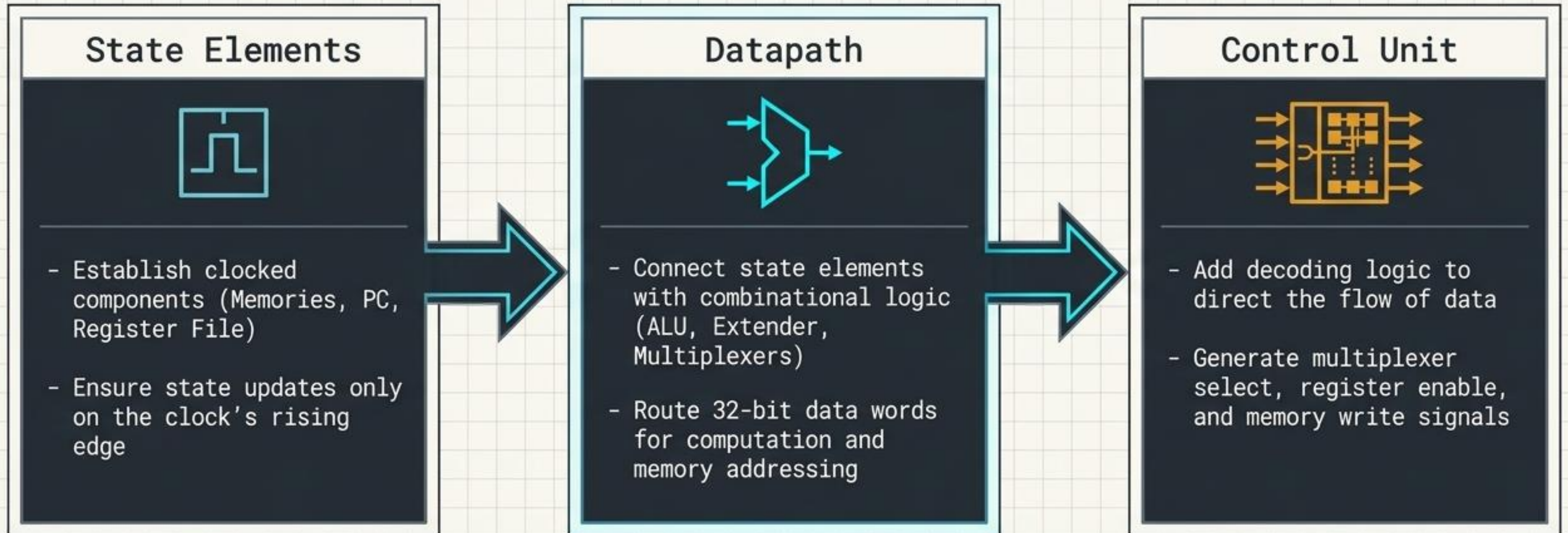
Defining the Microarchitecture Objective



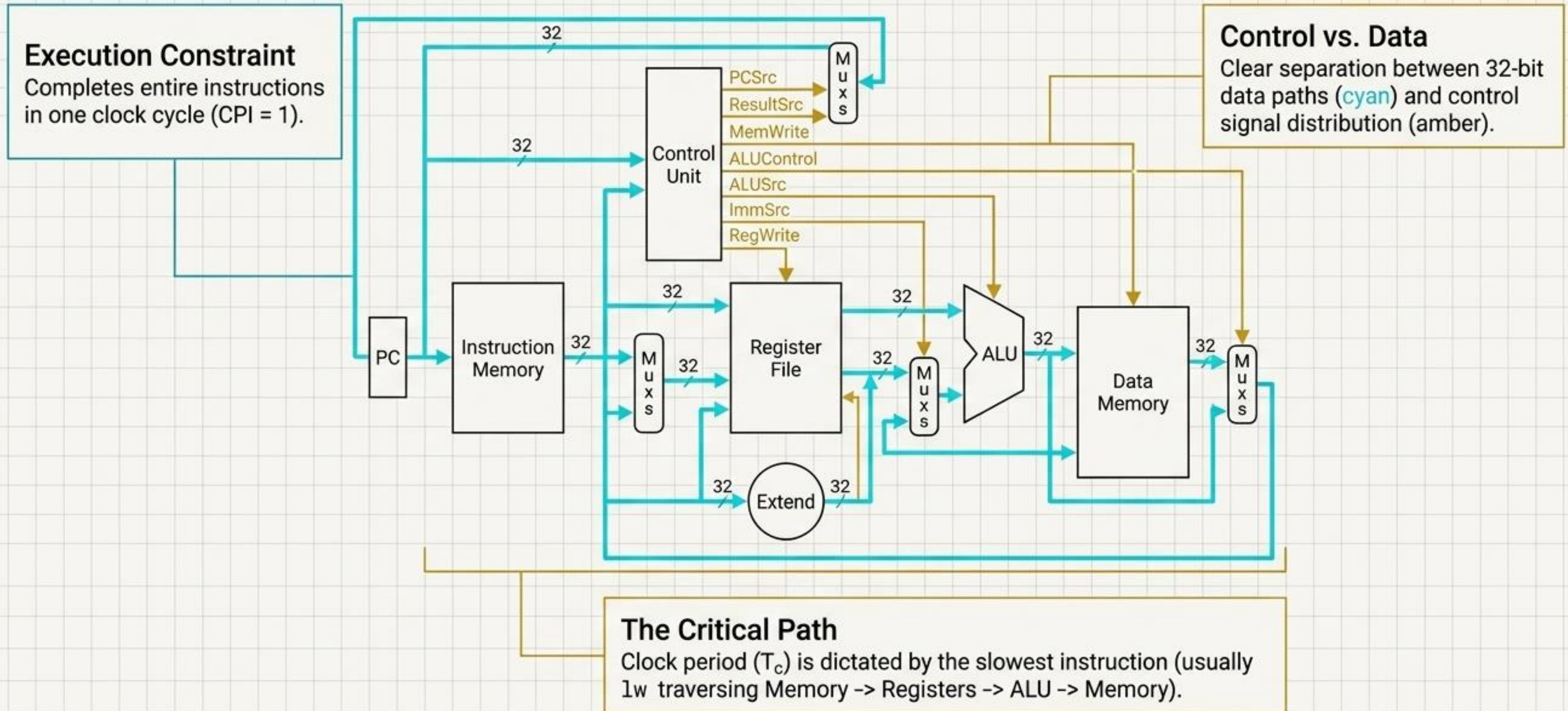
DEFINITION

Microarchitecture: The specific arrangement of registers, ALUs, finite state machines, and memories to physically implement the RISC-V instruction set.

The Three-Pillar Design Methodology



Top-Level Architecture: Single-Cycle Execution



Deconstructing the Hardware Building Blocks

Sequential (State)

(Clock-Synchronized)



- **Program Counter (PC)**: Holds current instruction address; updates on rising edge.
- **Register File**: 32-element x 32-bit storage; two read ports, one write port.

- **Memories**: Separate Instruction and Data memories to avoid structural hazards.


Combinational (Logic)

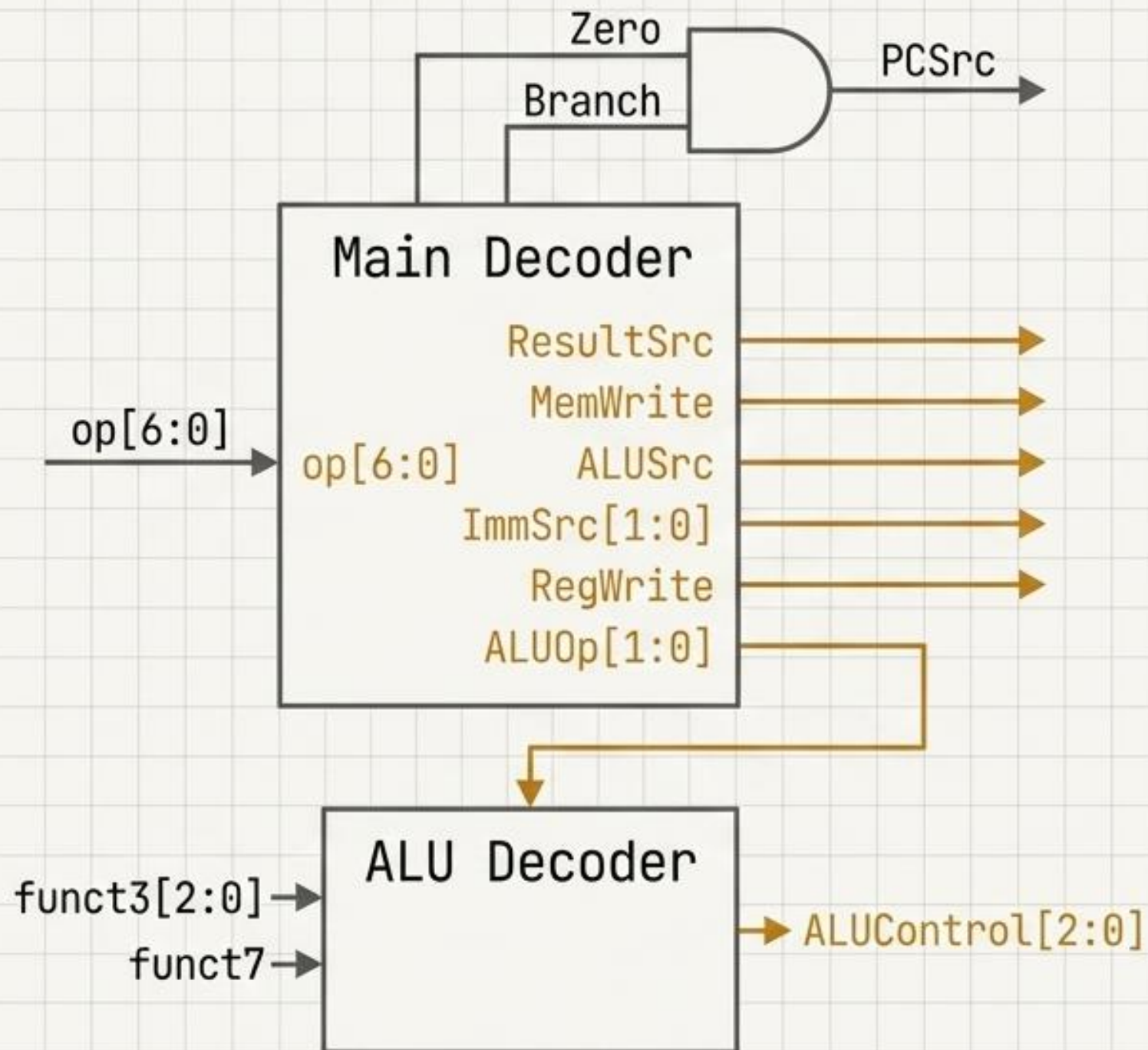
(Unclocked Logic)



- **ALU**: Executes arithmetic/logic via 3-bit control signals (e.g., add, sub, and).
- **Multiplexers**: Hardware routing switches (e.g., selecting PCNext or ALUSrc).

- **Extend Unit**: Sign-extends 12-bit instruction immediate fields to 32 bits.


Control Unit: Decoding Instructions into Actions



Instruction Splitting

Reads opcode (bits 6:0), funct3 (14:12), and funct7 (30).

Main Decoder

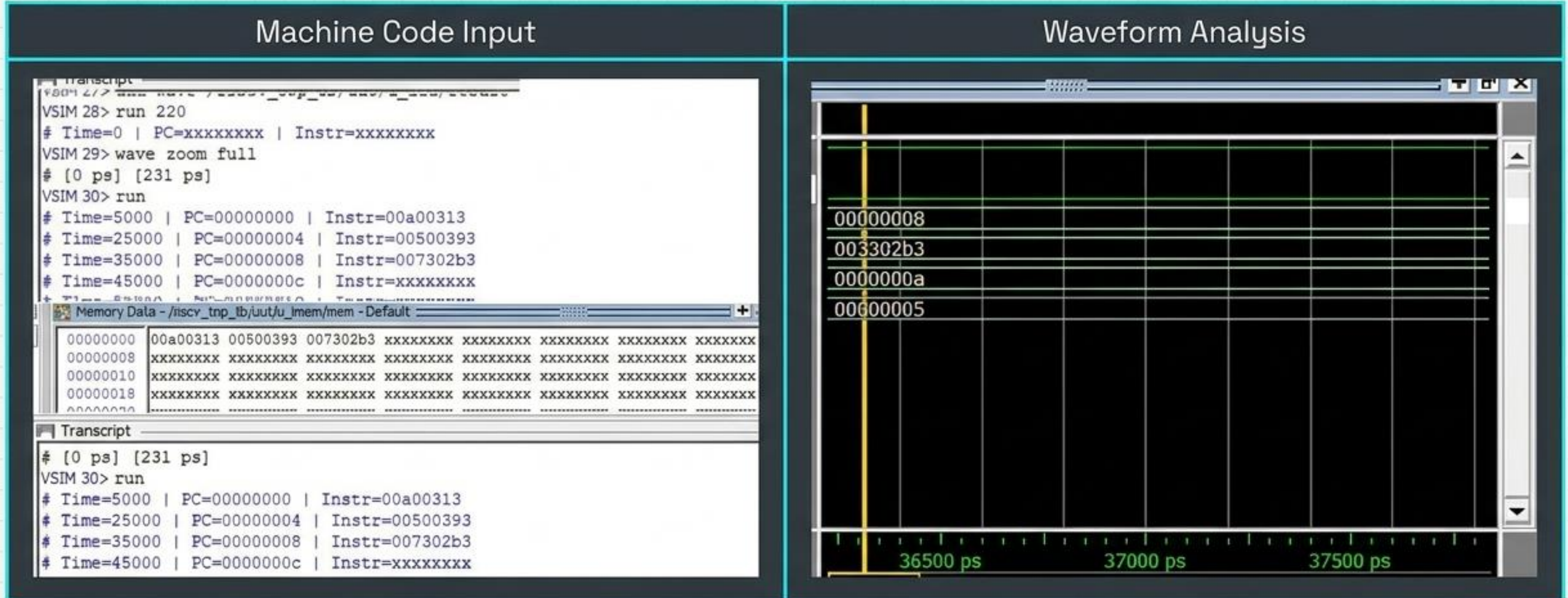
Identifies instruction type from the opcode to assert primary signals (MemWrite, RegWrite, Branch, ALUSrc).

ALU Decoder

Uses ALUOp alongside funct3 to generate the precise ALUControl signal for execution (e.g., 010 for AND, 001 for SUB).

Physical Verification: Simulating the Processor

Evidence Panel



Successful execution confirmed. Program counter, register states, and memory outputs perfectly match the expected behavioral simulation.

Conclusion & Engineering Lessons Learned

Project Achievements

- Successfully built and verified a functional single-cycle RV32I datapath and control unit.
- Bridged the gap between high-level RISC-V assembly and gate-level physical implementation.

Architecture Limitations

- The Bottleneck: Single-cycle execution means the clock frequency is strictly limited by the slowest instruction (Load Word).
- Highlights the necessity for advanced microarchitectures (Pipelining) for performance gains.

Core Insights

- Hardware/Software Interface: Demonstrated exactly how software instructions manipulate physical multiplexers and ALUs in parallel.